SE 216 – SOFTWARE PROJECT MANAGEMENT

REQUIREMENTS DOCUMENT

PROJECT NAME: **AI Personal Health & Fitness Coach**

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REQ. #	FUNCTIONAL REQUIREMENTS
1	- The system shall allow users to create an account and authenticate by logging in and logging out using an email address and password.
2	-The system shall display the daily AI-generated exercise plan and daily AI-generated meal plan on the home page after the user successfully logs in. The displayed plans shall include relevant details such as exercise names, sets, repetitions, intensity level, meal items, nutritional values, and visual illustrations where applicable.
3	-The system shall allow users to view detailed nutritional information (including calories and macronutrients) and allergen information (e.g., nuts, dairy, gluten) for each meal and each individual nutrient item where applicable..

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4	-The system shall adjust the intensity of the exercise plan based on the user's sleep data, calorie intake data, and previous exercise performance, which are collected from a smartwatch and phone motion sensors.
5	-The system shall display users' profile pictures alongside their names and scores on the leaderboard and allow filtering and sorting by region, performance metrics (total points, completed workouts,workout streak), and time period (daily, weekly, monthly)..
6	-The system shall generate a personalized workout plan using AI via the Hugging Face AI platform and shall display detailed visual illustrations for each exercise, including sets, repetitions, and proper form guidance
7	-The system shall create a personalized diet plan based on the user's dietary preferences (preferred foods, dietary restrictions), nutritional targets (calorie intake, macros,micros), health goals (weight loss, fitness, muscle gain), and physical characteristics (weight, height, body fat percentage, age).
8	-The system shall use the Hugging Face AI platform to perform research-based analysis and generate personalized recommendations for

	meals (calorie intake, nutrient balance, dietary restrictions), supplements, and exercises (correct form, sets, repetitions, weight), using recent verified scientific studies, user preferences, and user feedback
9	-The system shall allow users to rate their previous meals and exercises at the end of each day using numerical ratings and optional text comments. The collected feedback shall be analyzed using Hugging Face AI to improve future personalized recommendations.
10	-Users who rank within the top 10% on the leaderboard shall be awarded rewards such as badges or supplement discounts, which shall be automatically assigned once per month after the leaderboard is updated and displayed on both the leaderboard and the user’s profile page..
11	-The system shall allow users to customize their profile by updating personal information (name, profile picture, bio) and to add, remove, or organize their favorite meals and exercises.
12	-The system shall allow users to view full versions of the latest verified scientific papers and research about meals and exercises, as well as

	AI-generated summarized versions, with clear labeling of source and summary.
13	-The system shall provide medical warnings to users before they begin an exercise, based on their physical data, exercise type, and frequency.
14	-The system shall allow users to view other users' meal plans and exercise programs from their profiles only if the profile owner's privacy settings permit access.
15	-The system shall display users' favorite meals and exercises prominently on their profile and use them to personalize AI-generated recommendations for future meals and exercise plans.
16	-The system shall calculate user rankings on the leaderboard based on workout density (total completed workouts over a defined period) and streaks (consecutive days of activity).
REQ. #	**NON-FUNCTIONAL REQUIREMENTS**

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1	-The system shall maintain 99% uptime availability.
2	-The system shall use the Argon2 algorithm for password hashing and AES-256 encryption for storing personal data (name, birth date, profile information), ensuring compliance with KVKK regulations.
3	-The system shall ensure that users' meal plans and exercise programs are only visible to other users if the profile owner has explicitly allowed access, protecting sensitive dietary and exercise information.
4	-The app should work on both iOS (version 13.0 and above) and Android (version 8.0 and above).
5	-First time users should be able to sign in and login within 3 minutes, receive and apply exercise guidance within 5 minutes.
6	-Nutritional, allergen, and other displayed data shall be accurate and based on verified scientific sources to the best extent possible, and shall be regularly reviewed to ensure consistency and reliability.

7	-The system shall use OAuth 2.0 for authentication in login and registration processes.
8	-User sessions shall automatically expire after 15 minutes of inactivity.
9	-The system shall communicate via RESTful APIs.
10	-AI analysis of feedback shall be completed within 5 seconds to update recommendations.